Awareness, Usage, and Perspectives on ChatGPT in Dental Education Among Graduate and Undergraduate Students

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Abstract

This study aims to assess the knowledge, attitudes, and practices (KAP) related to ChatGPT among undergraduate students and graduates in Karachi, Pakistan, and to analyze an association with sociodemographic factors. A cross-sectional study was conducted over three months using a self-administered, structured questionnaire distributed via online platforms (WhatsApp, and emails). The questionnaire included four sections: demographics, knowledge, attitude, and practice related to ChatGPT. A non-probability convenience sampling technique was employed. Data were analyzed using SPSS version 26. Descriptive statistics summarized participant characteristics. Multiple linear regression was used to identify associations between KAP outcomes and sociodemographic variables. A p-value < 0.05 was considered statistically significant. A total of 347 participants responded, of whom 74.6% were female and 76.4% reported awareness of ChatGPT. Age, designation, and years of experience were significantly associated with knowledge ($p = 0.012$, 0.001, and 0.001, respectively). Attitude was significantly influenced by age and gender ($p = 0.03$ and 0.04, respectively), while only gender showed a significant association with perception ($p = 0.021$). Most respondents considered ChatGPT moderately accurate (54.8%) and

acknowledged its usefulness in education and research. Ethical concerns and uncertainty about ChatGPT replacing human roles were also noted. The majority of dental students and professionals in Karachi are aware of ChatGPT and hold a cautiously optimistic view of its role in dental education. While attitudes are generally positive, concerns remain regarding ethical implications and clinical applicability. Structured AI training and clear implementation guidelines are needed to support its effective integration in dental practice.